

Software Requirements Specification (SRS)

Project: *Production Planning System*

1. Introduction

The *Production Planning System* is a web-based application designed to streamline manufacturing operations by integrating product management, order processing, production scheduling, task assignment, and inventory tracking. It ensures efficient resource utilization, timely delivery, and transparent progress monitoring across all departments.

2. Objectives

- Simplify production planning and scheduling
- Enable role-based access for Admin, Manager, and Employee
- Automate task assignment and progress tracking
- Maintain real-time inventory updates
- Provide a unified dashboard for performance overview

3. Users

- **Admin:** System control, user management, and product setup
- **Manager:** Order creation, production planning, and task assignment
- **Employee:** Task execution and status updates

4. Functional Requirements

- User registration and login
- Role-based authentication and authorization
- Product CRUD operations
- Customer order creation and tracking

- Automatic production plan generation
- Task assignment and progress updates
- Inventory management and stock alerts
- Dashboard visualization of KPIs

5. Non-Functional Requirements

- **Security:** Spring Security + BCrypt password encryption
- **Performance:** Optimized REST APIs for fast data exchange
- **Usability:** Responsive UI with Bootstrap and React components
- **Maintainability:** Modular architecture using Spring Boot
- **Scalability:** Supports multiple departments and production lines

6. Technology Stack

Frontend	: React.js + Bootstrap
Backend	: Spring Boot + Spring Security
Database	: PostgreSQL
ORM	: Hibernate / Spring Data JPA
Build Tools	: Maven, npm
Version Control	: GitHub
Deployment	: Docker + AWS/Heroku

7. Database Tables

- users
- products
- orders

- production plans
- tasks
- employees
- inventory

8. Future Enhancements

- AI-based production forecasting
- Predictive inventory management
- Automated delay alerts via email/SMS
- Integration with IoT sensors for real-time tracking
- Advanced analytics dashboard